

Driver's real-time Drowsiness Detection using Adaptable Eye Aspect Ratio and Smart Alarm System

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Abstract— According to National Highway Traffic Safety Administration (NHTSA), drowsy driving leads to over 71,000 injuries, 1,500 deaths, and \$12.5 billion in monetary losses per year. The main issue that needs to be addressed is that the driver should be warned well before time to take precautionary measures. This paper suggests a radical solution to the problem by proposing a system that can detect the driver's drowsiness and can issue a timely warning. This paper describes a non-intrusive approach that takes into consideration the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) to quantify drowsiness by locating, monitoring, and analyzing both eyes and mouth, even in the dark. The real-time EAR is compared with the initial EAR of the drive and the MAR is compared with a set threshold of 20. The system alerts the user by issuing text and audio warnings. We also propose a smart alarm system that can be turned off by detecting a hand gesture. The data is collected from the user in real-time to test the system. This system predicted 9 out of 10 cases accurately, both in dark and light conditions, with different facial features including spectacles.

Keywords— *Adaptable Eye Aspect Ratio, Drowsiness detection, Facial landmarks, Hand gesture detection, Mouth Aspect Ratio, Sleepiness detection, Smart Alarm System*

I. INTRODUCTION

Drowsy driving or sleep-deprived driving occurs when a person operating the wheel is too tired to remain fully conscious and handle the wheel. This could lead to slower reaction timing and the driver might fall asleep. According to research, drowsy driving is riskier than texting while driving, distracted driving, or intoxicated driving combined. In 2017, 1,47,913 people were killed in road accidents, according to data released by the ministry of road transport and highways [1]. According to statistics, drivers with a reduced level of vigilance are responsible for 10% to 20% of all traffic incidents [2]. According to reports, 60 percent of adult drivers drove a car while drowsy in the year 2017-18. It is observed that young people aged 18 to 29 are the most vulnerable to sleep-related accidents [3]. Each year, the National Highway Traffic Safety Administration estimates that driver fatigue causes around 100,000 accidents [5]. One might face difficulty maintaining proper speed when drowsy driving. It is difficult to stay fully aware and the driver might slip into a brief state of unconsciousness.

Sleep deprivation accounts for the majority of cases of drowsy driving affecting the driver's ability to pay less attention to roads. This, drowsy driving may happen due to various reasons such as sleeping disorder, alcohol consumptions and the time of the day, etc. One of the most common symptoms of fatigue is the dropping of eyes. Sleepiness causes the eyelids to become heavier and thus droop. This might lead to the driver falling asleep completely. Some visible symptoms of drowsiness include frequent yawning, increased blinking rate, and difficulty concentrating. When a driver's exhaustion level increases, so do their ability to steer. As a result, the negative impact of fatigue on driving efficiency emphasizes the significance of early fatigue identification.

In recent times, computer vision-based diagnosis using image processing-based solutions have become increasingly popular. Research towards driver monitoring has gained momentum, especially for driver workload estimation [4], driver activity identification [5], secondary task identification [6], and driving style recognition [7]. Possible techniques for detecting driver's drowsiness can be divided into two categories:

- methods based on visual features such as eye blinking, yawning, and other physical changes [8].
- methods based on biological changes such as decrease in heart and pulse rate [9].

The critical issue that should be addressed, by a drowsiness detection system, is how early can it detect drowsiness and warn the driver to avoid accidents.

This paper focuses on providing a radical solution to this problem. We use CV tools to identify facial landmarks and then extract facial signals that indicate drowsiness. Since eyes are one the most salient features that reflect one's current state, we use eyelids' drooping and excessive blinking as one the factors to predict drowsiness. We do so by adaptable Eye Aspect Ratio (EAR). Moreover, we also use the movements of the lips to detect fatigue using the Mouth Aspect Ratio (MAR) and counting and keeping a count of frequent yawns. Thus, the goal of this paper is to propose a system that monitors human operator's vigilance and warns them the risk of accidents.

The rest of the paper is organized as follows: Section II describes the methodologies used to detect drowsiness. Section III provides a detailed explanation of our proposed drowsiness detection system and the alarm feature to alert the user. In section IV, we talk about the performance of our system under different circumstances and discuss the past work. Finally, we conclude the paper in section V stating some of the other applications of this system.

II. DISCUSSION

A. Facial Landmarks

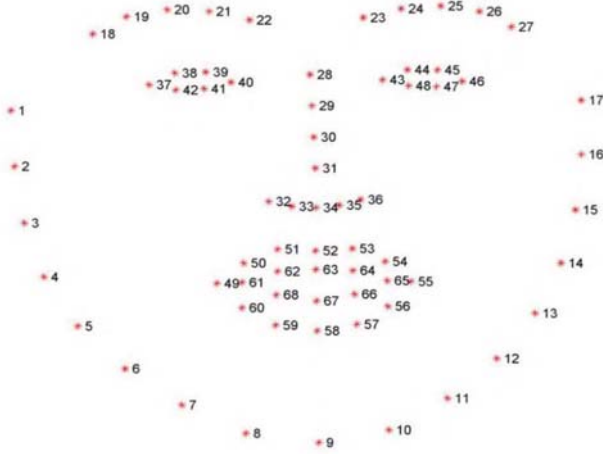


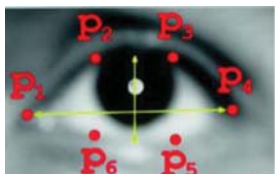
Figure 1: 68- points facial landmarks

Pre-existing features for facial landmark recognition are used to detect drowsiness and fatigue. To detect 68 facial landmarks, we used the open-source Python library d-lib. These predefined annotations help in shape prediction by recognizing various facial regions such as the eye-brows, eye, nose, and mouth region, among others (shown in Fig.1). Changes in the parameters of these distinct points represent the person's various expressions. [10]. The above coordinates represent the following-

Table I: Coordinates of various facial regions

Sr. no.	Region	Coordinates
1.	Mouth	[48, 68]
2.	Right eyebrow	[17, 22]
3.	Left eyebrow	[22, 27]
4.	Right eye	[36, 42]
5.	Left eye	[42, 48]
6.	Nose	[27, 35]
7.	Jaw	[0, 17]

B. Eye Aspect Ratio (EAR)



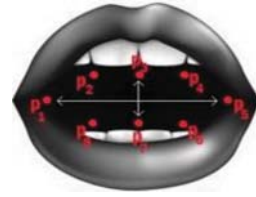
An aspect ratio describes the relationship of an object's width to its height [8]. The Eye Aspect Ratio is an estimate of the eye-opening state [11]. Each eye is represented by 6 (x, y)-coordinates (shown on the left), beginning from the left corner of an eye

and moving clockwise around the remaining of the points. This is a clever solution that calculates the ratio of distances of the sum of the difference between the vertical points to twice the difference between the horizontal points. EAR is always calculated for the left and right eye separately.

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}$$

Formula to calculate EAR

C. Mouth Aspect Ratio (MAR)



The Mouth Aspect Ratio is a measure of how open your mouth is. 8-coordinates are used to represent the mouth. Starting at the left corner of the mouth, the landmarks are marked in a clockwise direction around the rest of the region. The MAR is measured by dividing the vertical distance between the upper lip and lower lip by the horizontal distance between the corners of the lips [10]

$$MAR = \frac{\|p_2 - p_8\| + \|p_3 - p_7\| + \|p_4 - p_6\|}{2\|p_1 - p_5\|}$$

Formula to calculate MAR

III. PROPOSED METHODOLOGY

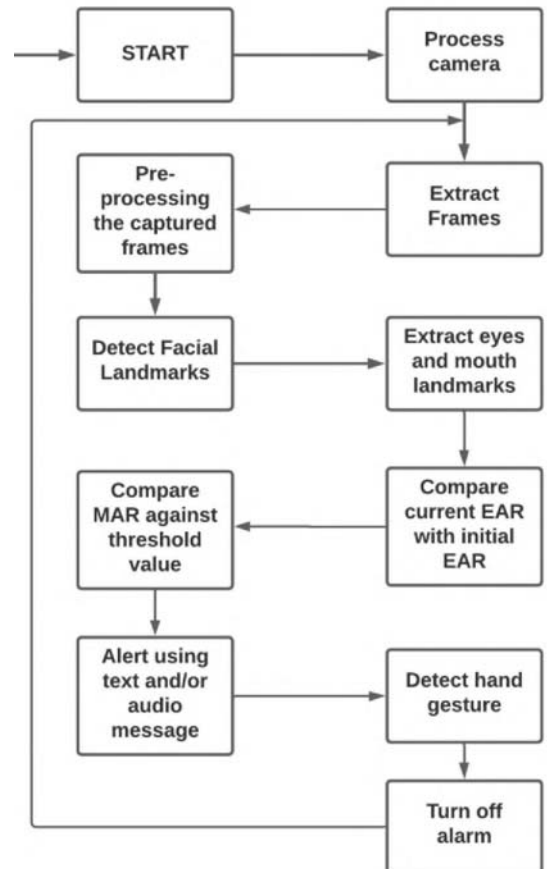


Figure 2: Block diagram

Fatigueness of the driver is one of the most casual factors for traffic accidents. Almost 1.2 million people lose their lives in traffic accidents every year around the world [1]. Thus, this paper proposes a non-intrusive system to detect and keep a track of driver's fatigue level based on its facial features. Fig. 2 outlines the proposed solution for driver's fatigue detection system.

A. Pre-processing:

To capture the video, we used a color camera with a 140-degree wide-angle attached to the car's dashboard at an angle that can see the driver's face. The video stream that is acquired by the camera is converted into frames so that it is easy to process and detect objects. Once the frames are read, they are resized and flipped across the y-axis (horizontal flipping). Then the system converts each frame from RGB to grayscale for easier processing.

These frames are then equalized to increase visibility in case of a dark environment. Histogram equalization distributes the pixels across the range. This is more visible in case of images with poor lighting and has little to no effect on images that have equal distribution of pixels (as shown in Fig. 3).

This system also applies Gaussian blur on each captured frame to remove background noise that might be because of the car movement or body movement. Background noise is more prevalent in case of dark background.

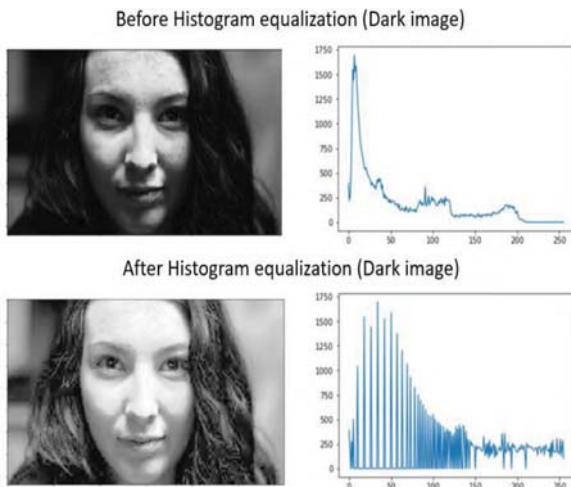


Figure 3: Histogram Equalization

B. Data gathering:

After processing the image such that the face area is completely visible, dlib's face detector is used to find and locate the face in the image. The pre-trained facial landmark detector in the dlib library is used to estimate the position of 68 (x, y)-coordinates (as shown in Fig. 1) that map to facial structures on the face to locate the facial landmarks. Following the location of the coordinates, the system establishes a bounding box around the face to concentrate properly.

Going ahead, we extract the coordinates of both eyes, and mouth and encapsulate the focused region into a polygon using a convex hull. We then draw contours around the eyes, and mouth region by joining the coordinates along the boundary for visual shape analysis (shown by green boundary in Fig. 5).

Subsequently, we calculated the initial Eye Aspect Ratio (EAR) of the driver as soon as the system is turned on, that is when the car is started and a face is detected. We do so because using a single standard EAR threshold value can result in inaccurate predictions for people with certain genetic disorders or people of various nationalities. Also, as there is a lack of datasets that contains image data for distinct eye sizes, nationalities and people who wear spectacles. As a result, comparing the driver's EAR to the database can result in an excessive number of false alarms.

C. Detection system:

Once the initial EAR of the driver is calculated and stored, we capture frames in groups of 30 to keep a tab on the fatigue level of the driver. EAR reduces when the driver shuts their eyes and increases back to the normal level when opened again (as shown in Fig. 4) [12]. We use this method to compute blinks.

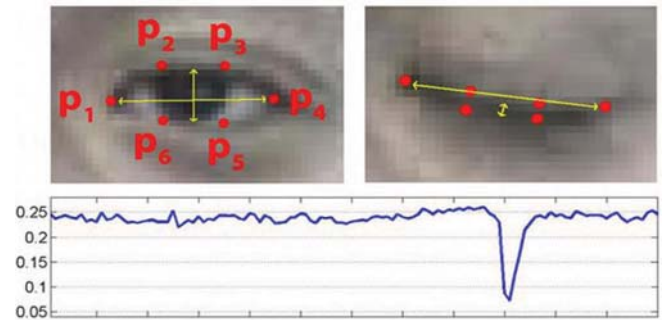


Figure 4: Change in EAR on opening and closing of eye

We compare real-time EAR to the initial EAR and display a text warning if we detect the driver's real-time EAR to be less than 20% of its initial EAR for more than 20 frames at a stretch. We consider this to be blinking (shown in Fig. 5). Being sleepy or drowsy tends to cause rapid blinking to stay alert.



Figure 5: Subject blinking

If the driver's eyes are shut more than 20% of that of its initial EAR for more than 48 frames, the alarm goes off and alerts the driver to take a break. The threshold for EAR is kept higher since we noticed that the EAR value does not go to 0 even when the eyes are completely shut (shown in fig. 5).

This system also keeps a tab of the number of yawns. It is not uncommon that one starts yawning when tired. The driver is considered yawning if the value of real-time Mouth aspect Ratio (MAR) is calculated to be above the threshold of 20, for more than 25 frames. If this is the case, a text warning is displayed on the screen (shown in Fig. 6).



Figure 6: Subject yawning

The MAR tends to increase when the mouth opens and decreases when the mouth is shut, making it easier to detect yawning (as shown in Fig. 7).

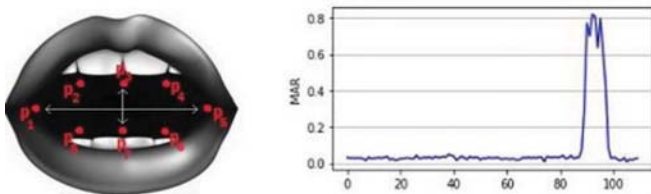


Figure 7: Change in MAR when mouth is shut vs open

This system gains an edge by locating facial coordinates and detecting drowsiness even in the dark (shown in Fig. 8). This feature helps to detect drowsiness in drivers during the nighttime when there is a greater chance of dozing off behind the wheel.



Figure 8: Facial coordinates' detection in dark

This process continues for as long as the system is working, that is till the car engine is not shut down.

D. Alarm system:

There are two ways in which this system alerts the driver: via text messages and by beeping an alarm sound. We have established the fact that the alarm only goes off when the user is found extremely drowsy, that is when the driver's eyes are shut more than 20% of that of its initial EAR for more than 48 frames. Now, to turn this alarm off, we have developed a smart alarm system that turns off the alarm based on hand gesture detection.

As soon as the alarm goes off, the system starts capturing frames to detect hand gestures by the driver, by finding a hand in Region of Interest (ROI) (shown in Fig. 9). It starts by capturing 30 frames from the live video stream, resizing and flipping them to capture hand gesture movements accurately. We then detect the shape of the object that is found in the ROI using Convex Hull. Next contours are drawn using the extreme points of the object as coordinates and the foreground is separated from the background making the hand detectable (shown in Fig. 9). On the condition that it is a hand that is detected by the system, the alarm stops beeping and the system continues to detect drowsiness of the driver.

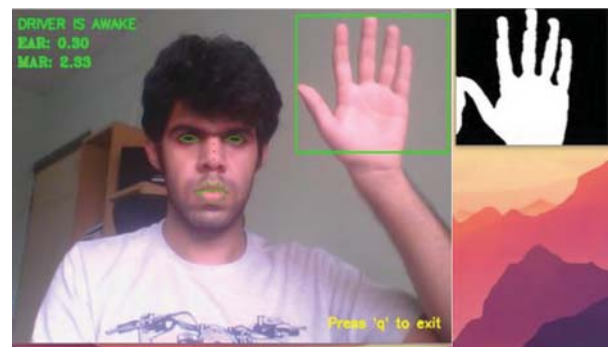


Figure 9: Hand gesture detection

The text warnings that appear on the screen, when the driver is yawning or blinking excessively, disappears on its own once the driver becomes alert or active.

IV. RESULT

A. Evaluation:

The proposed drowsiness detection system is embedded on a car dashboard to test it under different conditions. The frames extracted from the video stream are captured through a high-resolution camera that works during nighttime as well. We recorded the observations of 10 adults, all with different initial EAR (shown in table 2). We took into consideration the following parameters:

Time of day: Whether it is day or night, to check the efficiency of the system in dark.

Initial EAR: EAR recorded when the driver started driving.

Recorded (real-time) EAR: EAR recorded while driving.

Detected drowsiness: Prediction of the system (Nominal).

Actual drowsiness: Actual state of drowsiness (Nominal).

Table II shows the drowsiness detection for 10 subjects taken at different times of the day with different facial features such as a beard, spectacles, etc.

Table II- Drowsiness detection statistics

Sample	Time of Day	Ini. EAR	Rec. EAR	Detected Drowsy	Actual Drowsy
S1	Day	0.31	0.24	Yes	Yes
S2	Day	0.33	0.26	Yes	Yes
S3	Day	0.28	0.21	Yes	No
S4	Day	0.32	0.30	No	No
S5	Day	0.34	0.29	No	No
S6	Night	0.35	0.28	Yes	Yes
S7	Night	0.34	0.26	Yes	Yes
S8	Night	0.29	0.28	No	No
S9	Night	0.32	0.24	Yes	Yes
S10	Night	0.30	0.23	Yes	Yes

We notice that the system can plot facial landmarks and detect drowsiness even for the subjects that have a beard or wear spectacles (shown in Fig. 10) while driving with the same precision and pace as the one who does not have either.



Figure 10: Facial coordinates' detection with glasses

This system gains an edge by detecting drowsiness even when the face is at an angle that is not directly in front of the camera, provided maximum part of both eyes is visible (shown in Fig. 11).



Figure 11: Facial coordinates' detection from a side angle

The primary concern that needed to be verified for the efficient functioning of the system was whether it detects drowsiness with the same efficiency in dark as in daylight. We did that successfully.

We also recorded and observed their MAR and noticed that the system accurately predicts yawning in all the cases where the real-time MAR crosses the threshold. The average time that the system takes to detect drowsiness is 7-10 seconds which serves our aim of alerting the user well before time, before he asleep, to avoid any consequences. The system predicted 9 out of 10 cases accurately with respect to EAR.

B. Related work:

Regarding related work, one very recent study, conducted in 2020, for detecting drowsiness presented by Rateb Jabbar et al. [13] proposes a system using neural-network-based methodologies using multi-layer perceptron to detect microsleep symptoms. On an average, 83% of accuracy was achieved in all of the categories.

Caio Bezerra SoutoMaiores, Márcio José das ChagasMoura, João Mateus MarquesSantana, Isis DidierLinsab [12] aimed to implement a methodology for drowsiness detection using multilayer perceptron, random forest, and support vector machines (SVM) based on eye patterns of users monitored by video streams. Their SVM turned up to have the overall best performance with an average test accuracy of 94.9% on a public database namely DROZY.

Federico Guede-Fernández, Mireya Fernández-Chimeno, Juan Ramos-Castro, And Miguel A. García-González [14] implemented a drowsiness detection method based on fluctuations in the respiratory signals. Their paper is based on the analysis of respiratory rate variability (RRV). It managed to achieve a specificity of 96.6% and a sensitivity of 90.3% on average. 20 subjects, aged between 20 and 60, comprising of men and women, participated in the study.

Umit Budak, Varun Bajaj, Yaman Akbulut, Orhan Atila, Abdulkadir Sengur [15] suggested a method where they use EEG signals, spectral entropy and instantaneous frequency to calculate zero-crossing distribution features. They also incorporated deep feature extraction using pre-trained AlexNet and VGGNet on EEG spectrogram images. They observed this model on MIT-BIH Polysomnographic database and obtained an average accuracy score of 94.31%.

Wen-Bing Horng, Chih-Yuan Chen, Yi Chang, Chun-Hai Fan [16] proposed a real-time driver fatigue detection system that uses skin colour to locate the regions of eyes and then convert these images to the HSI model to distinguish pixels of eyeballs to determine whether the eyes are open or closed. The average precision rate of the system was 89.9%.

Qian Cheng, Wuhong Wang, Xiaobei Jiang, Shanyi Hou, Yong Qin [17] put forward a comprehensive method the pattern of the driver's facial landmark sequences shifts. The following variables were extracted from videos of 21 participants' faces taken during the experiment: EAR, MAR, PERCLOS, blink rate, statistics of blink duration, closing speed, reopening speed, and number of yawns. They used

four machine learning algorithms, including logistic regression algorithm which gave an accuracy of 85.7%.

This paper stands out because it presents a novel method for tracking both eye and mouth patterns, and it demonstrates the ability to work effectively even at night. This system issues a warning in advance to avoid any consequences by using a smart alarm system. This method excels due to its adaptable eye aspect ratio feature, which works well for people of all ages, nationalities, and genders.

V. CONCLUSION

This paper presented a non-intrusive system using computer vision to predict the drowsiness of the driver behind the wheel in real-time based on the frames extracted from the live video stream. When we found the driver sleepy or drowsy, or when we detected blinking or yawning, we issued a timely warning via text and/or audio, which helped us avoid injuries. We tested the system on ten adults (both men and women) by attaching a camera to their car dashboard and found that it read correctly 9 out of 10 times. The adults who took part in the study were of various ages and nationalities, and they wore different types of eyewear. The observations were made at various times throughout the day and the system worked with equal efficiency. To improve productivity in online lectures and at work, this system can be installed on a laptop or a PC. Furthermore, it can be used in security cabins to keep guards alert to avoid injuries, robberies, or any other mishaps. We can add voice commands to turn off the alarm or start playing music when the driver is detected sleepy to improve the detection system's efficiency and ease of use. Moreover, we can also add a feature of face recognition where the system can detect the identity of the driver, if already exists, by verifying it with the database. This way the system will not have to take in the original EAR for every new person each time.

REFERENCES

- [1] Rahul Chhabra (July 30, 2019). 40% of highway accidents occur due to drivers dozing off. <https://www.financialexpress.com/india-news/40-of-highway-accidents-occur-due-to-drivers-dozing-off/1659901/>.
- [2] L.M. Bergasa, J. Nuevo, M.A. Sotelo, and M. Vazquez, "Real-time system for monitoring driver vigilance," in *Proc. Intelligent Vehicle Symp.*, Parma, Italy, 2004.
- [3] Dr. Sunita Chowdhary (March 13, 2018). Drowsy Driving. <https://www.medindia.net/patientinfo/drowsy-driving.htm>
- [4] Y. Xing, C. Lv, D. Cao, H. Wang, and Y. Zhao, "Driver workload estimation using a novel hybrid method of error reduction ratio causality and support vector machine," *Measurement*, vol. 114, pp. 390–397, Jan. 2018.
- [5] Y. Xing et al., "Identification and analysis of driver postures for in-vehicle driving activities and secondary tasks recognition," *IEEE Trans. Comput. Social Syst.*, vol. 5, no. 1, pp. 95–108, Mar. 2018.
- [6] Y. Zhao et al., "An orientation sensor-based head tracking system for driver behaviour monitoring," *Sensors*, vol. 17, no. 11, p. 2692, 2017.
- [7] C. M. Martinez, M. Heucke, F.-Y. Wang, B. Gao, and D. Cao, "Driving style recognition for intelligent vehicle control and advanced driver assistance: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 13, Mar. 2016.
- [8] Christensson, P. (2017, December 2). Aspect Ratio Definition. Retrieved 2021, Mar 9, from <https://techterms.com>
- [9] Sikander, Gulbadan, and Shahzad Anwar. "Driver fatigue detection systems: A review." *IEEE Transactions on Intelligent Transportation Systems* 20.6 (2018): 2339-2352.
- [10] Hitendra Garg. "Drowsiness Detection of a Driver using Conventional Computer Vision Application", 2020 International Conference on Power Electronics & IoT Applications in Renewable Energy and its Control (PARC), 2020
- [11] Gaby Abou Haidar, Hasan Moussawi, Georgio Abou Saad, Abbas Chalhoub. "Robotic Feeder for Disabled People (RFDLP)", 2019 Fifth International Conference on Advances in Biomedical Engineering (ICABME), 2019
- [12] Caio Bezerra Souto Maior, Márcio José das Chagas Moura, João Mateus Marques Santana, Isis Didier Lins. "Real-time classification for autonomous drowsiness detection using eye aspect ratio", *Expert Systems with Applications*, 2020
- [13] Jabbar, Rateb, et al. "Driver drowsiness detection model using convolutional neural networks techniques for android application." 2020 IEEE International Conference on Informatics, IoT, and Enabling Technologies (ICIoT). IEEE, 2020.
- [14] F. Guede-Fernández, M. Fernández-Chimeno, J. Ramos-Castro and M. A. García-González, "Driver Drowsiness Detection Based on Respiratory Signal Analysis," in *IEEE Access*, vol. 7, pp. 81826-81838, 2019, doi: 10.1109/ACCESS.2019.2924481.
- [15] U. Budak, V. Bajaj, Y. Akbulut, O. Atila and A. Sengur, "An Effective Hybrid Model for EEG-Based Drowsiness Detection," in *IEEE Sensors Journal*, vol. 19, no. 17, pp. 7624-7631, 1 Sept.1, 2019, doi: 10.1109/JSEN.2019.2917850.
- [16] Yang, Guosheng, Yingzi Lin, and Prabir Bhattacharya. "A driver fatigue recognition model based on information fusion and dynamic Bayesian network." *Information Sciences* 180.10 (2010): 1942-1954
- [17] Qian Cheng, Wuhong Wang, Xiaobei Jiang, Shanyi Hou, Yong Qin. "Assessment of Driver Mental Fatigue Using Facial Landmarks" , *IEEE Access*, 2019.